```
> check()
── Documenting ─────────────────────────────────────────────────────────────────
ℹ Updating NICHES documentation
ℹ Loading NICHES

── Building ────────────────────────────────────────────────────────────────────
Setting env vars:
  • CFLAGS    : -Wall -pedantic -fdiagnostics-color=always
• CXXFLAGS  : -Wall -pedantic -fdiagnostics-color=always
• CXX11FLAGS: -Wall -pedantic -fdiagnostics-color=always
• CXX14FLAGS: -Wall -pedantic -fdiagnostics-color=always
• CXX17FLAGS: -Wall -pedantic -fdiagnostics-color=always
• CXX20FLAGS: -Wall -pedantic -fdiagnostics-color=always
── R CMD build ─────────────────────────────────────────────────────────────────
✔  checking for file '/Users/msbr/GitHub/NICHESv2-msbr/DESCRIPTION' ...
─  preparing 'NICHES':
  ✔  checking DESCRIPTION meta-information ...
─  checking for LF line-endings in source and make files and shell scripts
─  checking for empty or unneeded directories
Removed empty directory 'NICHES/tests'
─  building 'NICHES_2.0.0.tar.gz'

── Checking ────────────────────────────────────────────────────────────────────
Setting env vars:
  • _R_CHECK_CRAN_INCOMING_REMOTE_               : FALSE
• _R_CHECK_CRAN_INCOMING_                       : FALSE
• _R_CHECK_FORCE_SUGGESTS_                      : FALSE
• _R_CHECK_PACKAGES_USED_IGNORE_UNUSED_IMPORTS_: FALSE
• NOT_CRAN                                      : true
── R CMD check ─────────────────────────────────────────────────────────────────
─  using log directory
'/private/var/folders/kr/4bpg506d0p99cdp0lycjh_zm0000gn/T/Rtmp4KbVa3/filef95d7dbd50dc/NICHES.Rcheck'
─  using R version 4.5.2 (2025-10-31)
─  using platform: aarch64-apple-darwin20
─  R was compiled by
Apple clang version 16.0.0 (clang-1600.0.26.6)
GNU Fortran (GCC) 14.2.0
─  running under: macOS Sonoma 14.6.1
─  using session charset: UTF-8
─  using options '--no-manual --as-cran' (2.6s)
✔  checking for file 'NICHES/DESCRIPTION' ...
─  this is package 'NICHES' version '2.0.0'
─  package encoding: UTF-8
✔  checking package namespace information ...
─  checking package dependencies ... INFO (673ms)
Imports includes 21 non-default packages.
Importing from so many packages makes the package vulnerable to any of
them becoming unavailable.  Move as many as possible to Suggests and
use conditionally.
✔  checking if this is a source package ...
✔  checking if there is a namespace
✔  checking for executable files (528ms)
✔  checking for hidden files and directories ...
W  checking for portable file names ...
Found the following files with non-portable file names:
  docs/articles/01 NICHES Spatial.html
docs/articles/02 NICHES Single.html
docs/articles/03 Rat Alveolus.html
docs/articles/04 Differential CellToCell.html
docs/articles/05 Differential SystemToCell.html
docs/articles/06 Pseudotime.html
docs/articles/07 Spatiotemporal NICHES.html
docs/articles/08 Interoperability - NICHES.html
docs/articles/09 System Effects of Aberrant Cells.html
markdown/Ligand Classification Draft 2025-02-03.html
markdown/Receptor Classification Draft 2025-02-3.html
markdown/Receptor RuleSetOrganization with Patterns.md
markdown/RuleSetOrganization with Patterns.html
markdown/RuleSetOrganization with Patterns.md
vignettes/01 NICHES Spatial.Rmd
vignettes/02 NICHES Single.Rmd
vignettes/03 Rat Alveolus.Rmd
vignettes/04 Differential CellToCell.Rmd
vignettes/05 Differential SystemToCell.Rmd
vignettes/06 Pseudotime.Rmd
vignettes/07 Spatiotemporal NICHES.Rmd
vignettes/08 Interoperability - NICHES.Rmd
vignettes/09 System Effects of Aberrant Cells.Rmd
```

```
working scripts MSBR/markdown to annotations.R
working scripts MSBR/mindr example.R
working scripts MSBR/rule set function test script.R
working scripts MSBR
These are not fully portable file names.
See section ‘Package structure’ in the ‘Writing R Extensions’ manual.
✔  checking for sufficient/correct file permissions
✔  checking whether package ‘NICHES’ can be installed (2s)
✔  checking installed package size ...
✔  checking package directory
✔  checking for future file timestamps ...
N  checking DESCRIPTION meta-information ...
Malformed Description field: should contain one or more complete sentences.
N  checking top-level files
File
LICENSE
is not mentioned in the DESCRIPTION file.
Non-standard files/directories found at top level:
  ‘CLAUDE.md’ ‘_config.yml’ ‘_pkgdown.yml’ ‘docs’ ‘legacy-tests’
‘markdown’ ‘omnipathr-log’ ‘packageUpdateScript.R’
‘pre-knit-vignettes’ ‘working scripts MSBR’
✔  checking for left-over files ...
✔  checking index information
✔  checking package subdirectories ...
W  checking code files for non-ASCII characters ...
Found the following files with non-ASCII characters:
  R/CircuitPlot.R
R/NetworkPlot.R
R/RunCellToCell.R
R/RunCellToCellSpatial.R
R/RunCellToNeighborhood.R
R/RunCellToSystem.R
R/RunNeighborhoodToCell.R
R/RunSystemToCell.R
Portable packages must use only ASCII characters in their R code and
NAMESPACE directives, except perhaps in comments.
Use \uxxxx escapes for other characters.
Function ‘tools::showNonASCIIfile’ can help in finding non-ASCII
characters in files.
✔  checking R files for syntax errors ...
✔  checking whether the package can be loaded ...
✔  checking whether the package can be loaded with stated dependencies ...
✔  checking whether the package can be unloaded cleanly ...
✔  checking whether the namespace can be loaded with stated dependencies ...
✔  checking whether the namespace can be unloaded cleanly ...
W  checking dependencies in R code (3.3s)
'::' or ':::' imports not declared from:
  ‘ComplexHeatmap’ ‘SeuratObject’ ‘ggnewscale’ ‘netCoin’ ‘spNNGP’
‘viridis’
'library' or 'require' calls not declared from:
  ‘ComplexHeatmap’ ‘RColorBrewer’ ‘ggplot2’ ‘tidyverse’
'library' or 'require' calls in package code:
  ‘ComplexHeatmap’ ‘RColorBrewer’ ‘ggplot2’ ‘tidyverse’
Please use :: or requireNamespace() instead.
See section 'Suggested packages' in the 'Writing R Extensions' manual.
Namespaces in Imports field not imported from:
  ‘BisRNA’ ‘ggrepel’ ‘gtools’ ‘permute’ ‘plotrix’ ‘tibble’
All declared Imports should be used.
':::' call which should be '::': ‘Seurat:::PercentAbove’
See the note in ?`:::` about the use of this operator.
Unexported objects imported by ':::' calls:
  ‘spNNGP:::mk.n.indx.list’ ‘spNNGP:::mkNNIndxCB’
See the note in ?`:::` about the use of this operator.
✔  checking S3 generic/method consistency ...
✔  checking replacement functions ...
✔  checking foreign function calls ...
N  checking R code for possible problems (3.6s)
AggregateEdgeData: no visible global function definition for
‘aggregate’
ComputeEdgelist: no visible global function definition for ‘setNames’
ComputeEdgelist: no visible global function definition for ‘na.omit’
CustomHeatmapDendrogram: no visible global function definition for
‘colorRampPalette’
CustomHeatmapDendrogram: no visible global function definition for
‘rowAnnotation’
CustomHeatmapDendrogram: no visible global function definition for
‘anno_mark’
CustomHeatmapDendrogram: no visible global function definition for
‘gpar’
CustomHeatmapDendrogram: no visible global function definition for
```

‘Heatmap’
CustomHeatmapDendrogram: no visible global function definition for
‘unit’
CustomHeatmapDendrogram: no visible binding for global variable
‘colors.inferno’
CustomHeatmapDendrogram: no visible global function definition for
‘draw’
CustomHeatmapDendrogram: no visible binding for '<<-' assignment to
‘row.order.output’
CustomHeatmapDendrogram: no visible global function definition for
‘row_order’
NetworkPlot: no visible global function definition for ‘ggplot’
NetworkPlot: no visible global function definition for ‘aes_q’
NetworkPlot: no visible global function definition for ‘sym’
NetworkPlot: no visible global function definition for ‘theme_classic’
NetworkPlot: no visible global function definition for ‘geom_point’
NetworkPlot: no visible global function definition for
‘scale_alpha_continuous’
NetworkPlot: no visible global function definition for
‘scale_colour_gradientn’
NetworkPlot: no visible global function definition for ‘geom_segment’
NetworkPlot: no visible global function definition for ‘aes’
NetworkPlot: no visible binding for global variable
‘sending.barcode.umap1’
NetworkPlot: no visible binding for global variable
‘sending.barcode.umap2’
NetworkPlot: no visible binding for global variable
‘receiving.barcode.umap1’
NetworkPlot: no visible binding for global variable
‘receiving.barcode.umap2’
NetworkPlot: no visible binding for global variable
‘connectivity.to.plot’
NetworkPlot: no visible global function definition for ‘unit’
NetworkPlot: no visible global function definition for ‘ggtitle’
NetworkPlot: no visible global function definition for ‘NoLegend’
NetworkPlot: no visible global function definition for ‘aes_’
NetworkPlot: no visible global function definition for
‘scale_color_manual’
RunCellToCell: no visible global function definition for
‘return_celltypes’
ggCircuit: no visible global function definition for ‘ggplot’
ggCircuit: no visible global function definition for ‘aes’
ggCircuit: no visible binding for global variable ‘x’
ggCircuit: no visible binding for global variable ‘y’
ggCircuit: no visible global function definition for ‘geom_segment’
ggCircuit: no visible binding for global variable ‘x.start.offset’
ggCircuit: no visible binding for global variable ‘y.start.offset’
ggCircuit: no visible binding for global variable ‘x.end.offset’
ggCircuit: no visible binding for global variable ‘y.end.offset’
ggCircuit: no visible binding for global variable ‘feature.value’
ggCircuit: no visible global function definition for ‘unit’
ggCircuit: no visible global function definition for ‘geom_curve’
ggCircuit: no visible binding for global variable ‘x.start’
ggCircuit: no visible binding for global variable ‘y.start’
ggCircuit: no visible binding for global variable ‘x.end’
ggCircuit: no visible binding for global variable ‘y.end’
ggCircuit: no visible global function definition for
‘scale_alpha_continuous’
ggCircuit: no visible global function definition for ‘geom_point’
ggCircuit: no visible binding for global variable ‘system.fraction’
ggCircuit: no visible binding for global variable ‘node.label’
ggCircuit: no visible global function definition for
‘scale_color_manual’
ggCircuit: no visible global function definition for
‘scale_size_continuous’
ggCircuit: no visible global function definition for ‘theme_classic’
ggCircuit: no visible global function definition for ‘xlim’
ggCircuit: no visible global function definition for ‘ylim’
ggCircuit: no visible global function definition for ‘theme’
ggCircuit: no visible global function definition for ‘element_text’
gg_color_hue: no visible global function definition for ‘hcl’
Undefined global functions or variables:
  Heatmap NoLegend aes aes_ aes_q aggregate anno_mark colorRampPalette
colors.inferno connectivity.to.plot draw element_text feature.value
geom_curve geom_point geom_segment ggplot ggtitle gpar hcl na.omit
node.label receiving.barcode.umap1 receiving.barcode.umap2
return_celltypes rowAnnotation row_order scale_alpha_continuous
scale_color_manual scale_colour_gradientn scale_size_continuous
sending.barcode.umap1 sending.barcode.umap2 setNames sym
system.fraction theme theme_classic unit x x.end x.end.offset x.start

```
x.start.offset xlim y y.end y.end.offset y.start y.start.offset ylim
Consider adding
importFrom("grDevices", "colorRampPalette", "hcl")
importFrom("stats", "aggregate", "na.omit", "setNames")
to your NAMESPACE file.
✔  checking Rd files ...
✔  checking Rd metadata ...
✔  checking Rd line widths ...
✔  checking Rd cross-references ...
✔  checking for missing documentation entries ...
✔  checking for code/documentation mismatches (682ms)
W  checking Rd \usage sections ...
Undocumented arguments in Rd file 'CircuitPlot.Rd'
‘...’

Undocumented arguments in Rd file 'ComputeEdgelist.Rd'
‘nn.method’
Documented arguments not in \usage in Rd file 'ComputeEdgelist.Rd':
  ‘nn.methods’

Undocumented arguments in Rd file 'ggCircuit.Rd'
‘edge.scale.factor’ ‘arrow.head.angle’ ‘arrow.head.length’
‘autocrine.arrow.curvature’ ‘cols.use’ ‘edge.fixed.size’ ‘split.by’
‘min.edge.value’ ‘max.edge.value’ ‘title’

Functions with \usage entries need to have the appropriate \alias
entries, and all their arguments documented.
The \usage entries must correspond to syntactically valid R code.
See chapter ‘Writing R documentation files’ in the ‘Writing R
Extensions’ manual.
✔  checking Rd contents (336ms)
✔  checking for unstated dependencies in examples ...
✔  checking contents of ‘data’ directory ...
✔  checking data for non-ASCII characters ...
✔  checking LazyData
✔  checking data for ASCII and uncompressed saves ...
W  checking files in ‘vignettes’ ...
Files in the 'vignettes' directory but no files in 'inst/doc':
  ‘01 NICHES Spatial.Rmd’ ‘02 NICHES Single.Rmd’ ‘03 Rat Alveolus.Rmd’
‘04 Differential CellToCell.Rmd’ ‘05 Differential SystemToCell.Rmd’
‘06 Pseudotime.Rmd’ ‘07 Spatiotemporal NICHES.Rmd’ ‘08
Interoperability - NICHES.Rmd’ ‘09 System Effects of Aberrant
Cells.Rmd’ ‘data/pbmc3k_processed.h5ad’
‘omnipathr-log/omnipathr-20220527-0956.log’
‘omnipathr-log/omnipathr-20220619-2009.log’
Files named as vignettes but with no recognized vignette engine:
  ‘vignettes/01 NICHES Spatial.Rmd’
‘vignettes/02 NICHES Single.Rmd’
‘vignettes/03 Rat Alveolus.Rmd’
‘vignettes/04 Differential CellToCell.Rmd’
‘vignettes/05 Differential SystemToCell.Rmd’
‘vignettes/06 Pseudotime.Rmd’
‘vignettes/07 Spatiotemporal NICHES.Rmd’
‘vignettes/08 Interoperability - NICHES.Rmd’
‘vignettes/09 System Effects of Aberrant Cells.Rmd’
(Is a VignetteBuilder field missing?)
–  checking examples ... NONE
N  checking package vignettes
Package has ‘vignettes’ subdirectory but apparently no vignettes.
Perhaps the ‘VignetteBuilder’ information is missing from the
DESCRIPTION file?
  ✔  checking for non-standard things in the check directory
✔  checking for detritus in the temp directory

See
‘/private/var/folders/kr/4bpg506d0p99cdp0lycjh_zm0000gn/T/Rtmp4KbVa3/filef95d7dbd50dc/NICHES.Rcheck/00check.log’
for details.

── R CMD check results ──────────────────────────────────────── NICHES 2.0.0 ────
Duration: 20.9s

❯ checking for portable file names ... WARNING
Found the following files with non-portable file names:
  docs/articles/01 NICHES Spatial.html
docs/articles/02 NICHES Single.html
docs/articles/03 Rat Alveolus.html
docs/articles/04 Differential CellToCell.html
docs/articles/05 Differential SystemToCell.html
docs/articles/06 Pseudotime.html
docs/articles/07 Spatiotemporal NICHES.html
```

```
docs/articles/08 Interoperability - NICHES.html
docs/articles/09 System Effects of Aberrant Cells.html
markdown/Ligand Classification Draft 2025-02-03.html
markdown/Receptor Classification Draft 2025-02-3.html
markdown/Receptor RuleSetOrganization with Patterns.md
markdown/RuleSetOrganization with Patterns.html
markdown/RuleSetOrganization with Patterns.md
vignettes/01 NICHES Spatial.Rmd
vignettes/02 NICHES Single.Rmd
vignettes/03 Rat Alveolus.Rmd
vignettes/04 Differential CellToCell.Rmd
vignettes/05 Differential SystemToCell.Rmd
vignettes/06 Pseudotime.Rmd
vignettes/07 Spatiotemporal NICHES.Rmd
vignettes/08 Interoperability - NICHES.Rmd
vignettes/09 System Effects of Aberrant Cells.Rmd
working scripts MSBR/markdown to annotations.R
working scripts MSBR/mindr example.R
working scripts MSBR/rule set function test script.R
working scripts MSBR
These are not fully portable file names.
See section ‘Package structure’ in the ‘Writing R Extensions’ manual.

❯ checking code files for non-ASCII characters ... WARNING
Found the following files with non-ASCII characters:
  R/CircuitPlot.R
R/NetworkPlot.R
R/RunCellToCell.R
R/RunCellToCellSpatial.R
R/RunCellToNeighborhood.R
R/RunCellToSystem.R
R/RunNeighborhoodToCell.R
R/RunSystemToCell.R
Portable packages must use only ASCII characters in their R code and
NAMESPACE directives, except perhaps in comments.
Use \uxxxx escapes for other characters.
Function ‘tools::showNonASCIIfile’ can help in finding non-ASCII
characters in files.

❯ checking dependencies in R code ... WARNING
'::' or ':::' imports not declared from:
  ‘ComplexHeatmap’ ‘SeuratObject’ ‘ggnewscale’ ‘netCoin’ ‘spNNGP’
‘viridis’
'library' or 'require' calls not declared from:
  ‘ComplexHeatmap’ ‘RColorBrewer’ ‘ggplot2’ ‘tidyverse’
'library' or 'require' calls in package code:
  ‘ComplexHeatmap’ ‘RColorBrewer’ ‘ggplot2’ ‘tidyverse’
Please use :: or requireNamespace() instead.
See section 'Suggested packages' in the 'Writing R Extensions' manual.
Namespaces in Imports field not imported from:
  ‘BisRNA’ ‘ggrepel’ ‘gtools’ ‘permute’ ‘plotrix’ ‘tibble’
All declared Imports should be used.
':::' call which should be '::': ‘Seurat:::PercentAbove’
See the note in ?`:::` about the use of this operator.
Unexported objects imported by ':::' calls:
  ‘spNNGP:::mk.n.indx.list’ ‘spNNGP:::mkNNIndxCB’
See the note in ?`:::` about the use of this operator.

❯ checking Rd \usage sections ... WARNING
Undocumented arguments in Rd file 'CircuitPlot.Rd'
‘...’

Undocumented arguments in Rd file 'ComputeEdgelist.Rd'
‘nn.method’
Documented arguments not in \usage in Rd file 'ComputeEdgelist.Rd':
  ‘nn.methods’

Undocumented arguments in Rd file 'ggCircuit.Rd'
‘edge.scale.factor’ ‘arrow.head.angle’ ‘arrow.head.length’
‘autocrine.arrow.curvature’ ‘cols.use’ ‘edge.fixed.size’ ‘split.by’
‘min.edge.value’ ‘max.edge.value’ ‘title’

Functions with \usage entries need to have the appropriate \alias
entries, and all their arguments documented.
The \usage entries must correspond to syntactically valid R code.
See chapter ‘Writing R documentation files’ in the ‘Writing R
Extensions’ manual.

❯ checking files in ‘vignettes’ ... WARNING
Files in the 'vignettes' directory but no files in 'inst/doc':
```

```
  ‘01 NICHES Spatial.Rmd’ ‘02 NICHES Single.Rmd’ ‘03 Rat Alveolus.Rmd’
‘04 Differential CellToCell.Rmd’ ‘05 Differential SystemToCell.Rmd’
‘06 Pseudotime.Rmd’ ‘07 Spatiotemporal NICHES.Rmd’ ‘08
Interoperability - NICHES.Rmd’ ‘09 System Effects of Aberrant
Cells.Rmd’ ‘data/pbmc3k_processed.h5ad’
‘omnipathr-log/omnipathr-20220527-0956.log’
‘omnipathr-log/omnipathr-20220619-2009.log’
Files named as vignettes but with no recognized vignette engine:
  ‘vignettes/01 NICHES Spatial.Rmd’
‘vignettes/02 NICHES Single.Rmd’
‘vignettes/03 Rat Alveolus.Rmd’
‘vignettes/04 Differential CellToCell.Rmd’
‘vignettes/05 Differential SystemToCell.Rmd’
‘vignettes/06 Pseudotime.Rmd’
‘vignettes/07 Spatiotemporal NICHES.Rmd’
‘vignettes/08 Interoperability - NICHES.Rmd’
‘vignettes/09 System Effects of Aberrant Cells.Rmd’
(Is a VignetteBuilder field missing?)

❯ checking DESCRIPTION meta-information ... NOTE
Malformed Description field: should contain one or more complete sentences.

❯ checking top-level files ... NOTE
File
LICENSE
is not mentioned in the DESCRIPTION file.
Non-standard files/directories found at top level:
  ‘CLAUDE.md’ ‘_config.yml’ ‘_pkgdown.yml’ ‘docs’ ‘legacy-tests’
‘markdown’ ‘omnipathr-log’ ‘packageUpdateScript.R’
‘pre-knit-vignettes’ ‘working scripts MSBR’

❯ checking R code for possible problems ... NOTE
AggregateEdgeData: no visible global function definition for
‘aggregate’
ComputeEdgelist: no visible global function definition for ‘setNames’
ComputeEdgelist: no visible global function definition for ‘na.omit’
CustomHeatmapDendrogram: no visible global function definition for
‘colorRampPalette’
CustomHeatmapDendrogram: no visible global function definition for
‘rowAnnotation’
CustomHeatmapDendrogram: no visible global function definition for
‘anno_mark’
CustomHeatmapDendrogram: no visible global function definition for
‘gpar’
CustomHeatmapDendrogram: no visible global function definition for
‘Heatmap’
CustomHeatmapDendrogram: no visible global function definition for
‘unit’
CustomHeatmapDendrogram: no visible binding for global variable
‘colors.inferno’
CustomHeatmapDendrogram: no visible global function definition for
‘draw’
CustomHeatmapDendrogram: no visible binding for '<<-' assignment to
‘row.order.output’
CustomHeatmapDendrogram: no visible global function definition for
‘row_order’
NetworkPlot: no visible global function definition for ‘ggplot’
NetworkPlot: no visible global function definition for ‘aes_q’
NetworkPlot: no visible global function definition for ‘sym’
NetworkPlot: no visible global function definition for ‘theme_classic’
NetworkPlot: no visible global function definition for ‘geom_point’
NetworkPlot: no visible global function definition for
‘scale_alpha_continuous’
NetworkPlot: no visible global function definition for
‘scale_colour_gradientn’
NetworkPlot: no visible global function definition for ‘geom_segment’
NetworkPlot: no visible global function definition for ‘aes’
NetworkPlot: no visible binding for global variable
‘sending.barcode.umap1’
NetworkPlot: no visible binding for global variable
‘sending.barcode.umap2’
NetworkPlot: no visible binding for global variable
‘receiving.barcode.umap1’
NetworkPlot: no visible binding for global variable
‘receiving.barcode.umap2’
NetworkPlot: no visible binding for global variable
‘connectivity.to.plot’
NetworkPlot: no visible global function definition for ‘unit’
NetworkPlot: no visible global function definition for ‘ggtitle’
NetworkPlot: no visible global function definition for ‘NoLegend’
```

```
NetworkPlot: no visible global function definition for ‘aes_’
NetworkPlot: no visible global function definition for
‘scale_color_manual’
RunCellToCell: no visible global function definition for
‘return_celltypes’
ggCircuit: no visible global function definition for ‘ggplot’
ggCircuit: no visible global function definition for ‘aes’
ggCircuit: no visible binding for global variable ‘x’
ggCircuit: no visible binding for global variable ‘y’
ggCircuit: no visible global function definition for ‘geom_segment’
ggCircuit: no visible binding for global variable ‘x.start.offset’
ggCircuit: no visible binding for global variable ‘y.start.offset’
ggCircuit: no visible binding for global variable ‘x.end.offset’
ggCircuit: no visible binding for global variable ‘y.end.offset’
ggCircuit: no visible binding for global variable ‘feature.value’
ggCircuit: no visible global function definition for ‘unit’
ggCircuit: no visible global function definition for ‘geom_curve’
ggCircuit: no visible binding for global variable ‘x.start’
ggCircuit: no visible binding for global variable ‘y.start’
ggCircuit: no visible binding for global variable ‘x.end’
ggCircuit: no visible binding for global variable ‘y.end’
ggCircuit: no visible global function definition for
‘scale_alpha_continuous’
ggCircuit: no visible global function definition for ‘geom_point’
ggCircuit: no visible binding for global variable ‘system.fraction’
ggCircuit: no visible binding for global variable ‘node.label’
ggCircuit: no visible global function definition for
‘scale_color_manual’
ggCircuit: no visible global function definition for
‘scale_size_continuous’
ggCircuit: no visible global function definition for ‘theme_classic’
ggCircuit: no visible global function definition for ‘xlim’
ggCircuit: no visible global function definition for ‘ylim’
ggCircuit: no visible global function definition for ‘theme’
ggCircuit: no visible global function definition for ‘element_text’
gg_color_hue: no visible global function definition for ‘hcl’
Undefined global functions or variables:
  Heatmap NoLegend aes aes_ aes_q aggregate anno_mark colorRampPalette
colors.inferno connectivity.to.plot draw element_text feature.value
geom_curve geom_point geom_segment ggplot ggtitle gpar hcl na.omit
node.label receiving.barcode.umap1 receiving.barcode.umap2
return_celltypes rowAnnotation row_order scale_alpha_continuous
scale_color_manual scale_colour_gradientn scale_size_continuous
sending.barcode.umap1 sending.barcode.umap2 setNames sym
system.fraction theme theme_classic unit x x.end x.end.offset x.start
x.start.offset xlim y y.end y.end.offset y.start y.start.offset ylim
Consider adding
importFrom("grDevices", "colorRampPalette", "hcl")
importFrom("stats", "aggregate", "na.omit", "setNames")
to your NAMESPACE file.

❯ checking package vignettes ... NOTE
Package has ‘vignettes’ subdirectory but apparently no vignettes.
Perhaps the ‘VignetteBuilder’ information is missing from the
DESCRIPTION file?

  0 errors ✔ | 5 warnings ✖ | 4 notes ✖
```